

Hailin Wang

Ph.D. candidate at Tongji University & The Hong Kong Polytechnic University

 Shanghai, China
  hailin.wang@connect.polyu.hk
 176 2339 6448
  [hailin.wang](https://github.com/hailin.wang)
 [hailiin](#)
 [hailiin](#)

 0000-0001-5759-2455

About me

I am Hailin Wang, a passionate researcher and developer in the field of geotechnical engineering. My work focuses on the application of multiscale micromechanical methods to understand the behavior of geomaterials under various loading conditions. I have a strong interest in computational modeling, numerical simulations, and the integration of machine learning techniques into geotechnical analysis.

Education

- | | |
|---|---|
| BEng Huazhong University of Science and Technology , Civil Engineering | Wuhan, China
Sept 2016 – June 2020 |
| <ul style="list-style-type: none"> • Weighted grade: 91.79 (rank 1/64) • Honors: Received the National Scholarship and Merit Student of Huazhong University of Science and Technology for three consecutive years (2017-2019) | |
| PhD Tongji University , Geotechnical Engineering | Shanghai, China
Sept 2020 – present |
| <ul style="list-style-type: none"> • Advisor: Prof. Xiaoqiang Gu | |
| PhD The Hong Kong Polytechnic University , Geotechnical Engineering | Hong Kong, China
Sept 2021 – present |
| <ul style="list-style-type: none"> • Thesis: Multiscale Modelling of Granular Soils with Inherent and Induced Fabric Anisotropy • Advisor: Prof. Zhen-Yu Yin | |

Projects

- | | |
|--|---------------------|
| abqpy
Type Hints for Abaqus/Python Scripting <ul style="list-style-type: none"> • Enable intelligent type hinting for Abaqus/Python scripting • 300+ GitHub stars, 50+ forks | June 2022 – present |
|--|---------------------|

Publications

- | | |
|---|-----------|
| Uncertainty Quantification of Soil-Structure Interface Properties with an Enhanced Hypoplastic Interface Model
Hai-Lin Wang , Yin-Fu Jin, Zhen-Yu Yin, Xiao-Qiang Gu
(International Journal of Geomechanics) | June 2024 |
| Evaluation of Soil-Structure Interface Models
Hai-Lin Wang , Zhen-Yu Yin, Yin-Fu Jin, Xiao-Qiang Gu
10.1061/JGGEFK.GTENG-11486 (Journal of Geotechnical and Geoenvironmental Engineering) | July 2024 |
| Evaluation of Soil-Structure Interface Models Considering Cyclic Loading Effect
Hai-Lin Wang , Zhen-Yu Yin, Xiao-Qiang Gu, Yin-Fu Jin
10.1002/nag.3831 (International Journal for Numerical and Analytical Methods in Geomechanics) | Nov 2024 |

Integration of critical state theory into a micromechanical model for granular materials accounting for fabric evolution Hai-Lin Wang , Xiaoqiang Gu, Zhen-Yu Yin, Chao-Fa Zhao 10.1016/j.compgeo.2025.107379 (Computers and Geotechnics)	Oct 2025
A generalized anisotropic failure criterion for granular materials based on a multi-scale micromechanical approach Hai-Lin Wang , Xiaoqiang Gu, Zhen-Yu Yin 10.1016/j.ijsolstr.2025.113696 (International Journal of Solids and Structures)	Jan 2026
Micromechanics-inspired modeling of granular materials Hai-Lin Wang , Xiaoqiang Gu, Zhen-Yu Yin 10.1016/j.ijsolstr.2026.114263 (International Journal of Solids and Structures)	Oct 2026

Selected Honors

- National Scholarship (2017-2019)
- Merit Student of Huazhong University of Science and Technology (2017-2019)

Skills

Research Areas: Geotechnics, Geomechanics

Languages: Python, C++, Matlab, Fortran